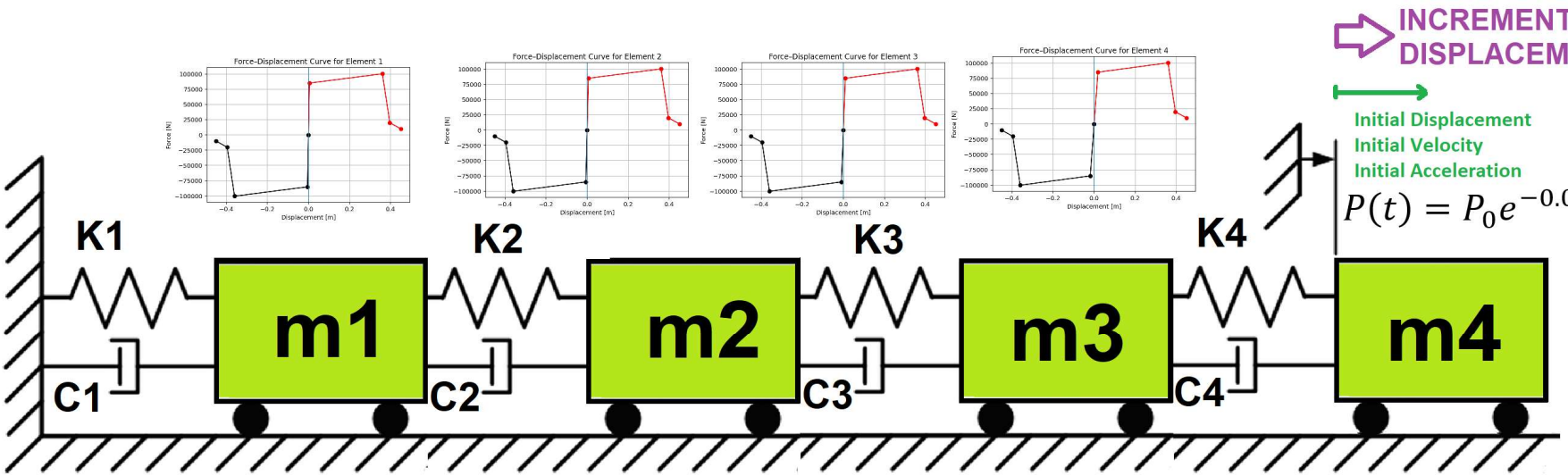
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

COMPREHENSIVE EVALUATION OF THE EQUIVALENT VISCOUS DAMPING RATIO OF A NONLINEAR MDOF SYSTEM UNDER MULTIPLE LOADING PROTOCOLS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

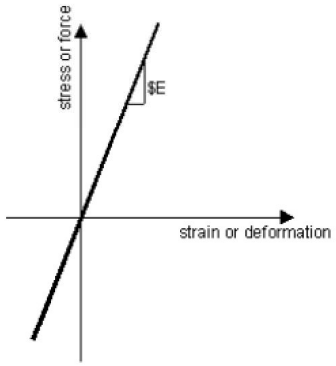
$\Delta_d = Lateral\ Displaement\ from\ Dynamic\ Analysis$

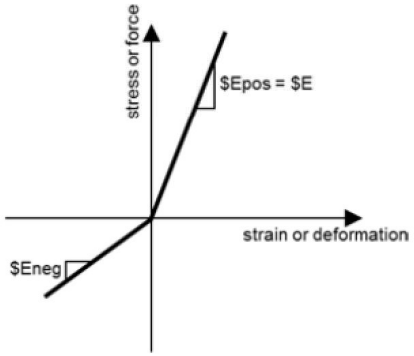
$\Delta_y = Lateral\ Yield\ Displaement\ from\ Pushover\ Analysis$

$\Delta_u = Lateral\ Ultimate\ Displaement\ from\ Pushover\ Analysis$

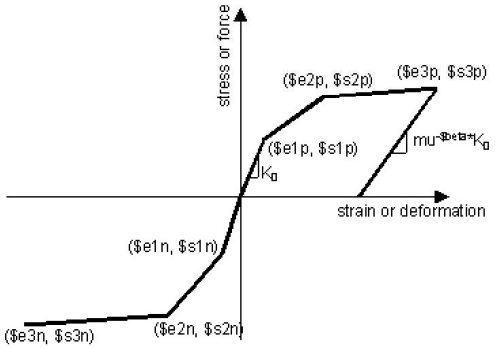
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

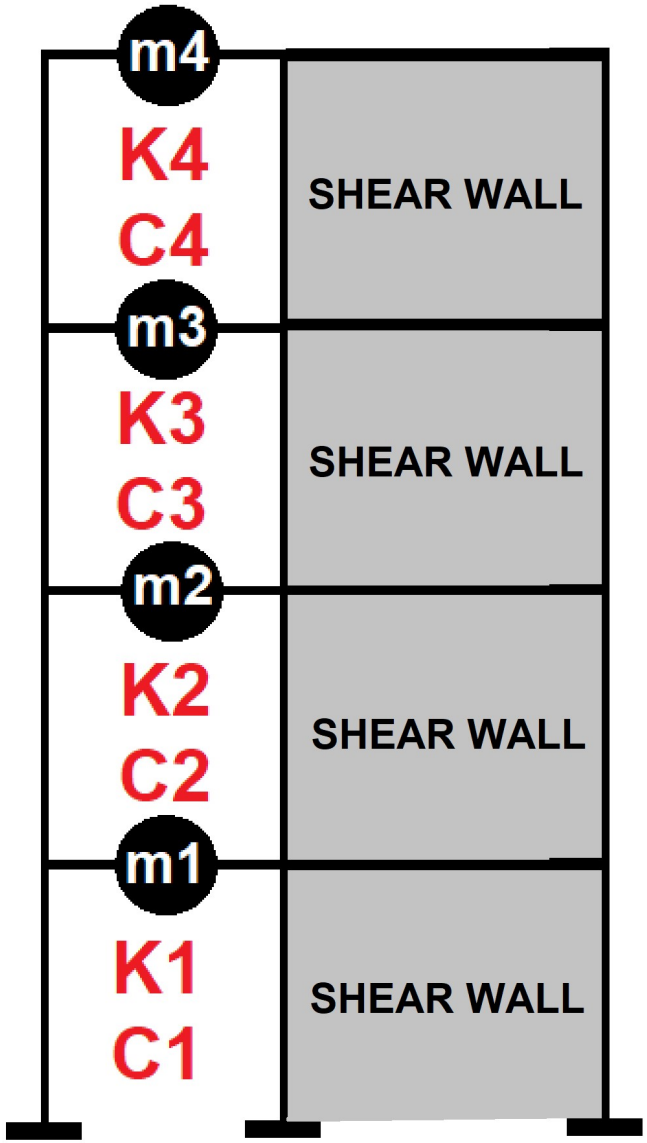




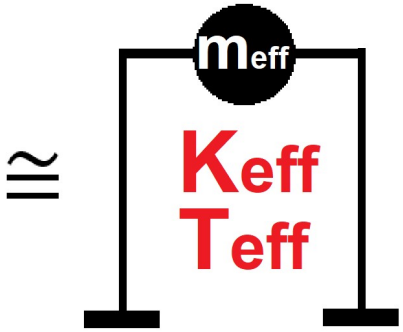
ELASTIC MATERIAL



INELASTIC MATERIAL

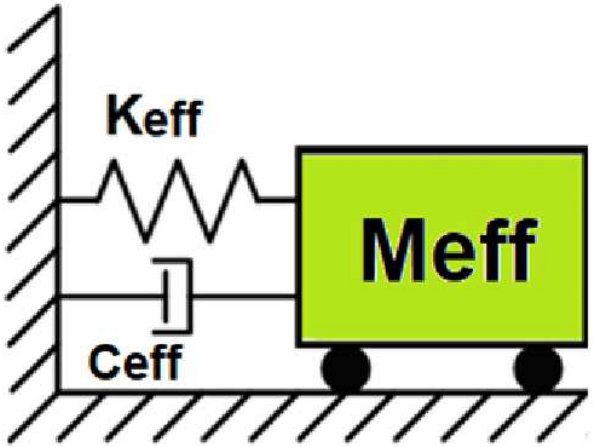


Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment



$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i} \qquad m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d} \qquad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

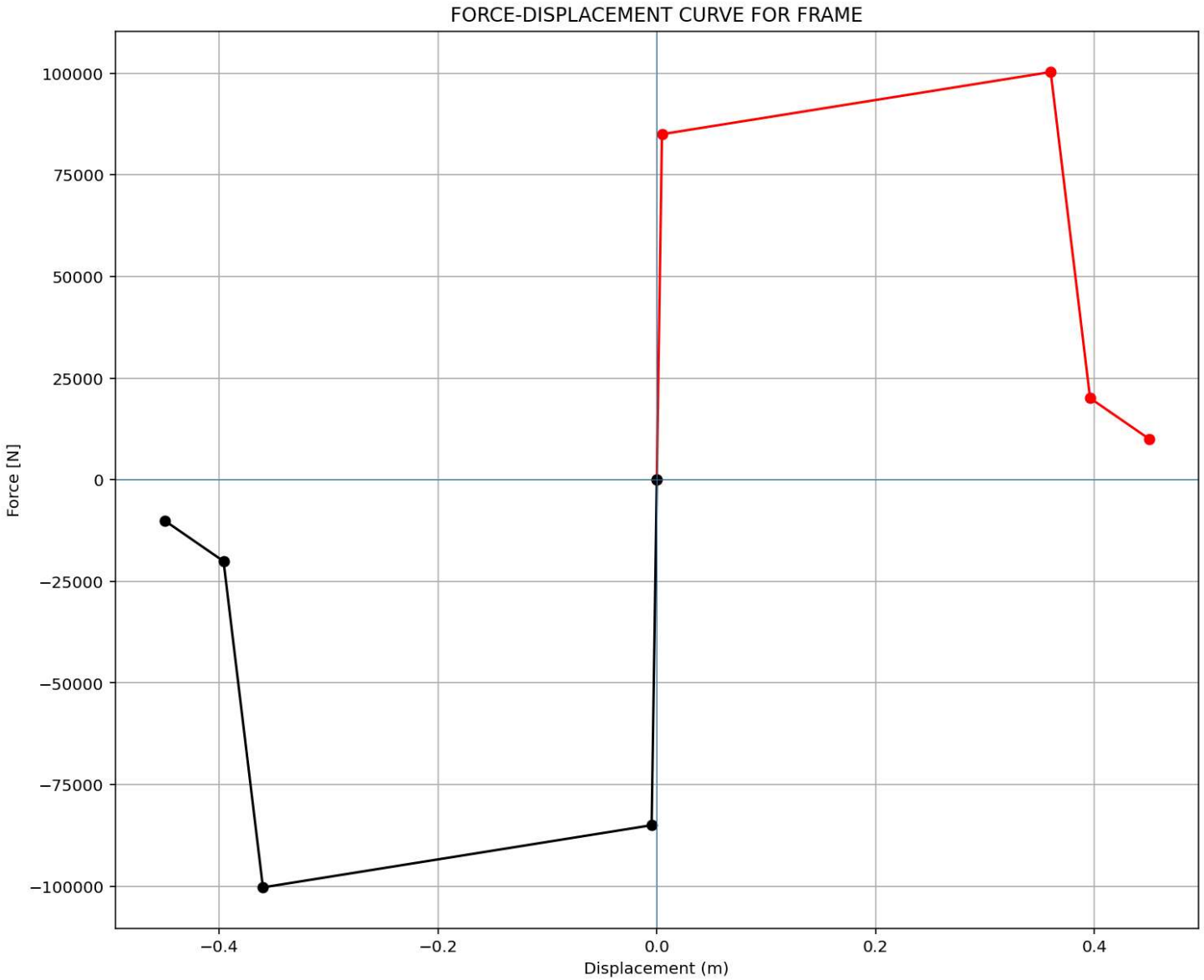
$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

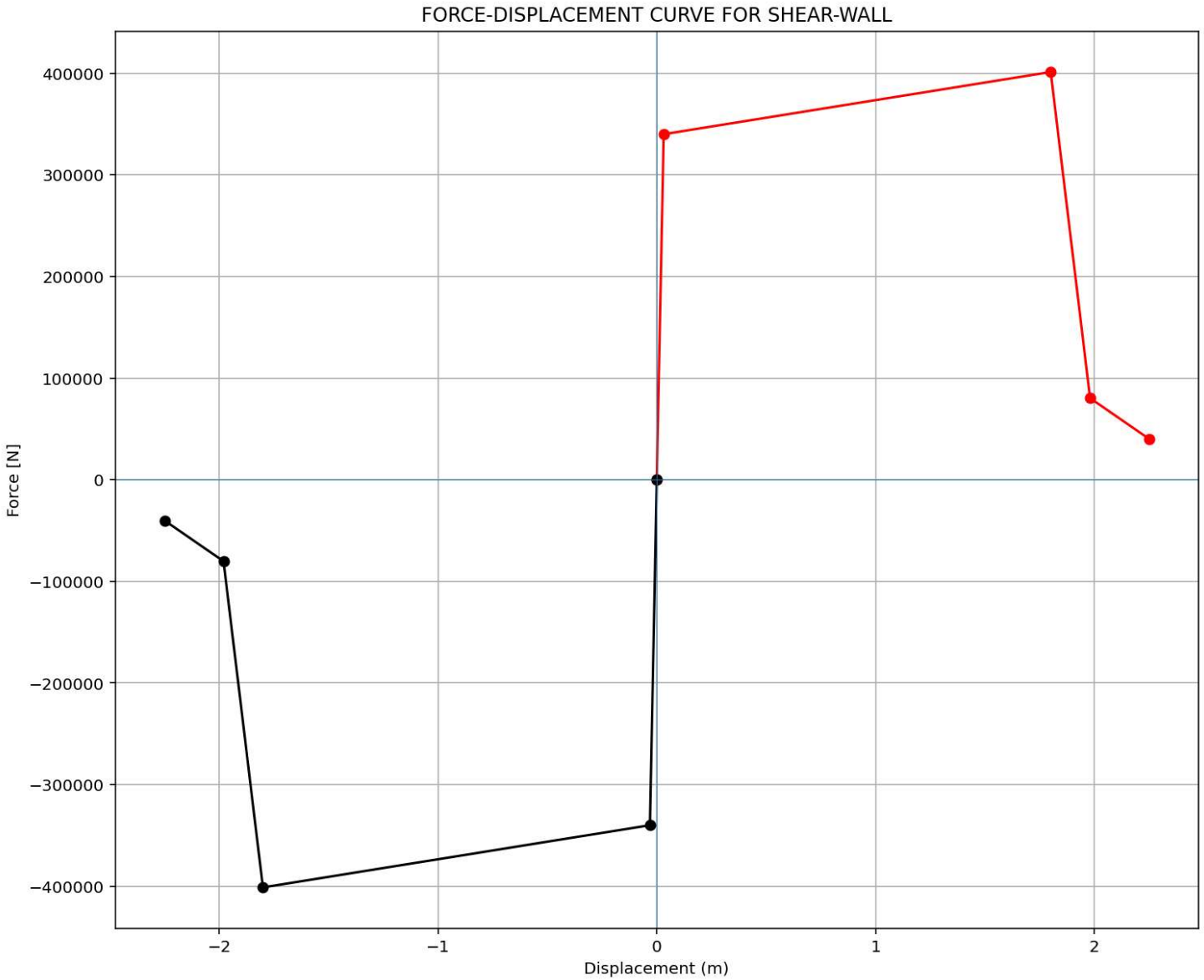
where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{\sec} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

The equivalent viscous damping ratio is then:

$$\xi_{eq} = \frac{100\, E_d}{4\pi\, E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.


FORCE-DISPLACEMENT CURVE FOR FRAME
Force [N]
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4
Displacement (m)


FORCE-DISPLACEMENT CURVE FOR SHEAR-WALL
Force [N]
400000
300000
200000
100000
0
−100000
−200000
−300000
−400000
−2
−1
0
1
2
Displacement (m)

Spyder (Python 3.12)

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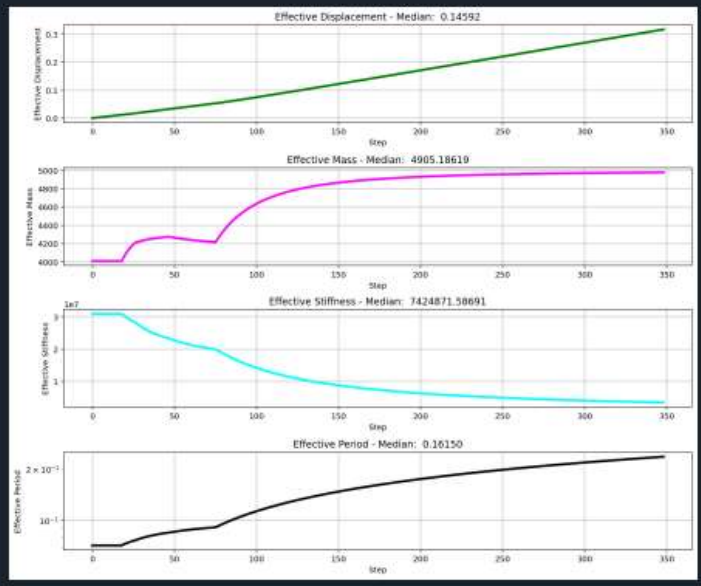
P(t)_MDOF_TWO_8_ANA.py | COMPUTE_EFFECTIVE_...ES_FUN_PUSHOVER.py

```python
#%% EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH P
# Change MDOF to SDOF System with Displacement Based Design Concept
# THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
"""
This script implements a displacement-based pushover transformation,
 converting a multi-degree-of-freedom (MDOF) system into an equivalent
 single-degree-of-freedom (SDOF) system for seismic assessment.
 It calculates effective modal properties—displacement, mass, and
 stiffness—by weighting element forces and nodal displacements according
 to a presumed deformed shape.
 The derived effective period provides a simplified dynamic characteristic
 for performance-based engineering. The visualizations effectively track the
 evolution of these equivalent parameters throughout the nonlinear analysis steps.
"""
#-----------------------------------------------------------------------
displacement_X_1 = np.array(list(node_displacements.values())[1])   # DOF 02
displacement_X_2 = np.array(list(node_displacements.values())[2])   # DOF 03
displacement_X_3 = np.array(list(node_displacements.values())[3])   # DOF 04
displacement_X_4 = np.array(list(node_displacements.values())[4])   # DOF 05

ele_force_01 = np.array(list(ele_force.values())[0])   # ELEMENT 01
ele_force_02 = np.array(list(ele_force.values())[1])   # ELEMENT 02
ele_force_03 = np.array(list(ele_force.values())[2])   # ELEMENT 03
ele_force_04 = np.array(list(ele_force.values())[3])   # ELEMENT 04

STIFF_X_1 = np.abs(ele_force_01 / displacement_X_1)
STIFF_X_2 = np.abs(ele_force_02 / displacement_X_2)
STIFF_X_3 = np.abs(ele_force_03 / displacement_X_3)
STIFF_X_4 = np.abs(ele_force_04 / displacement_X_4)

MX2 = (MASS[0] * np.square(displacement_X_1) +
       MASS[1] * np.square(displacement_X_2) +
       MASS[2] * np.square(displacement_X_3) +
       MASS[3] * np.square(displacement_X_4))
```



26 %

IPython Console | Files | Help | Variable Explorer | Debugger | Plots | History

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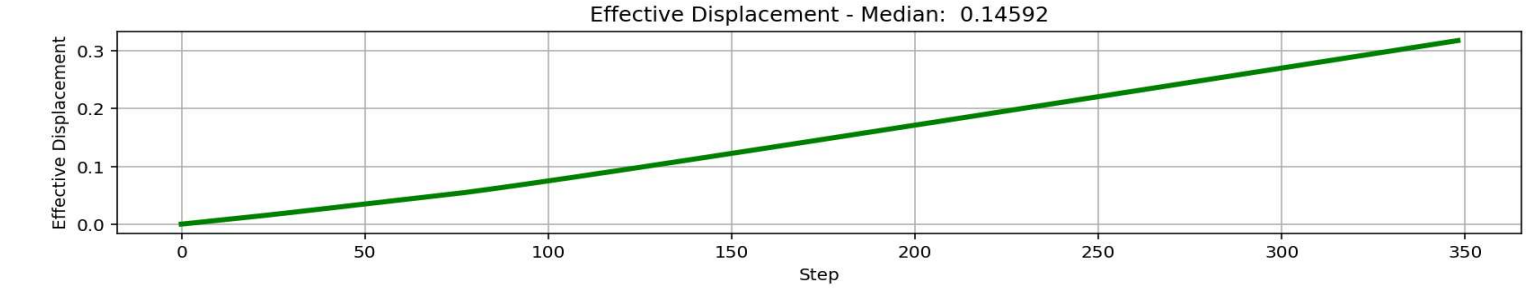
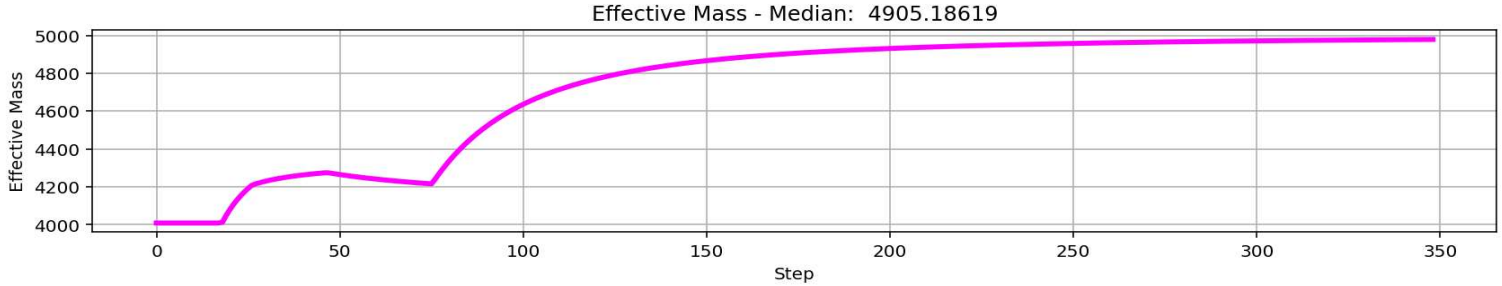
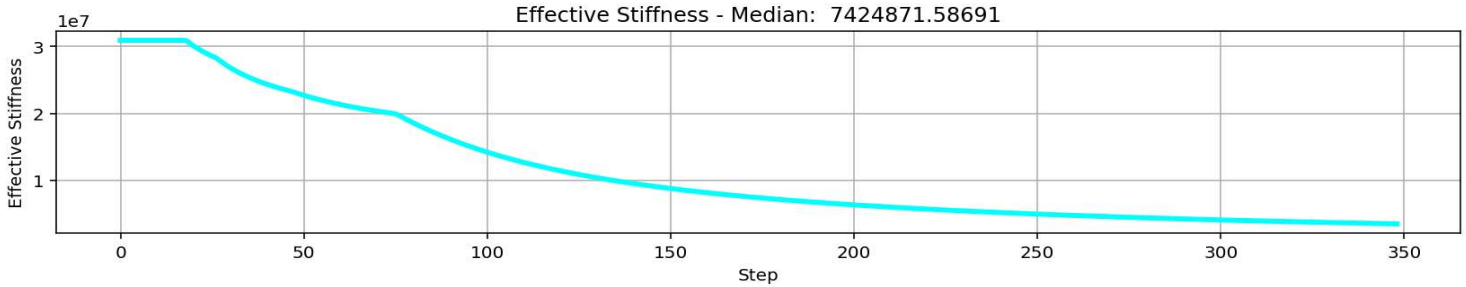


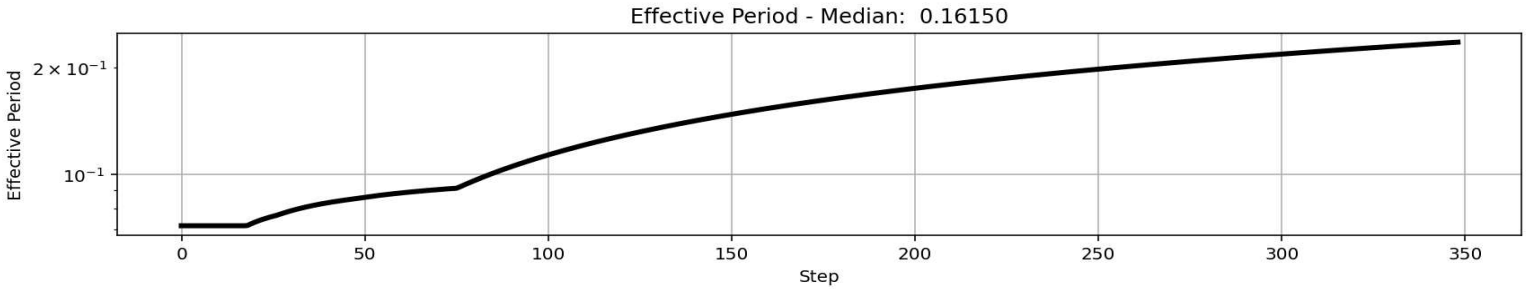
$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i}$$



$$m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

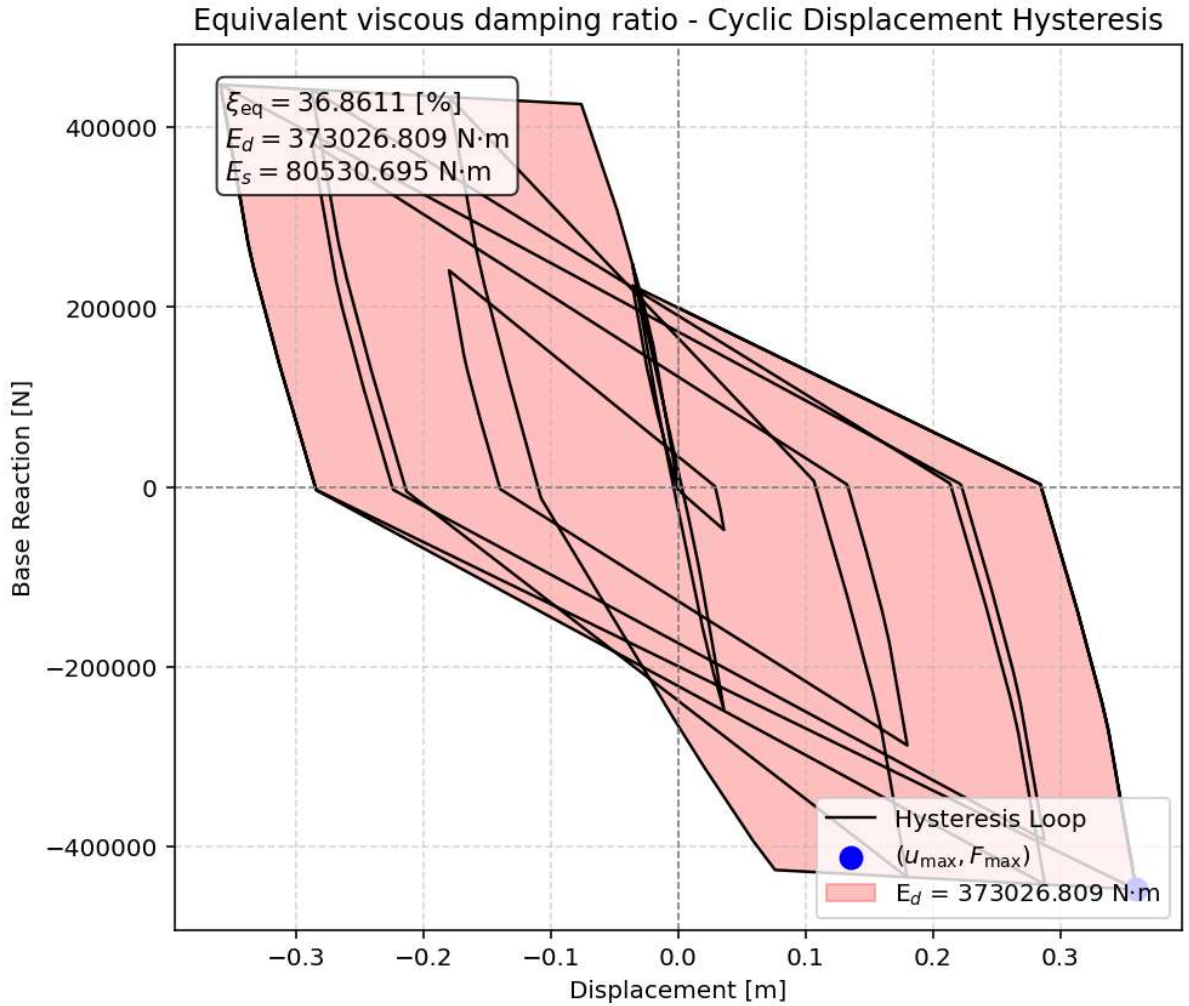


$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d}$$

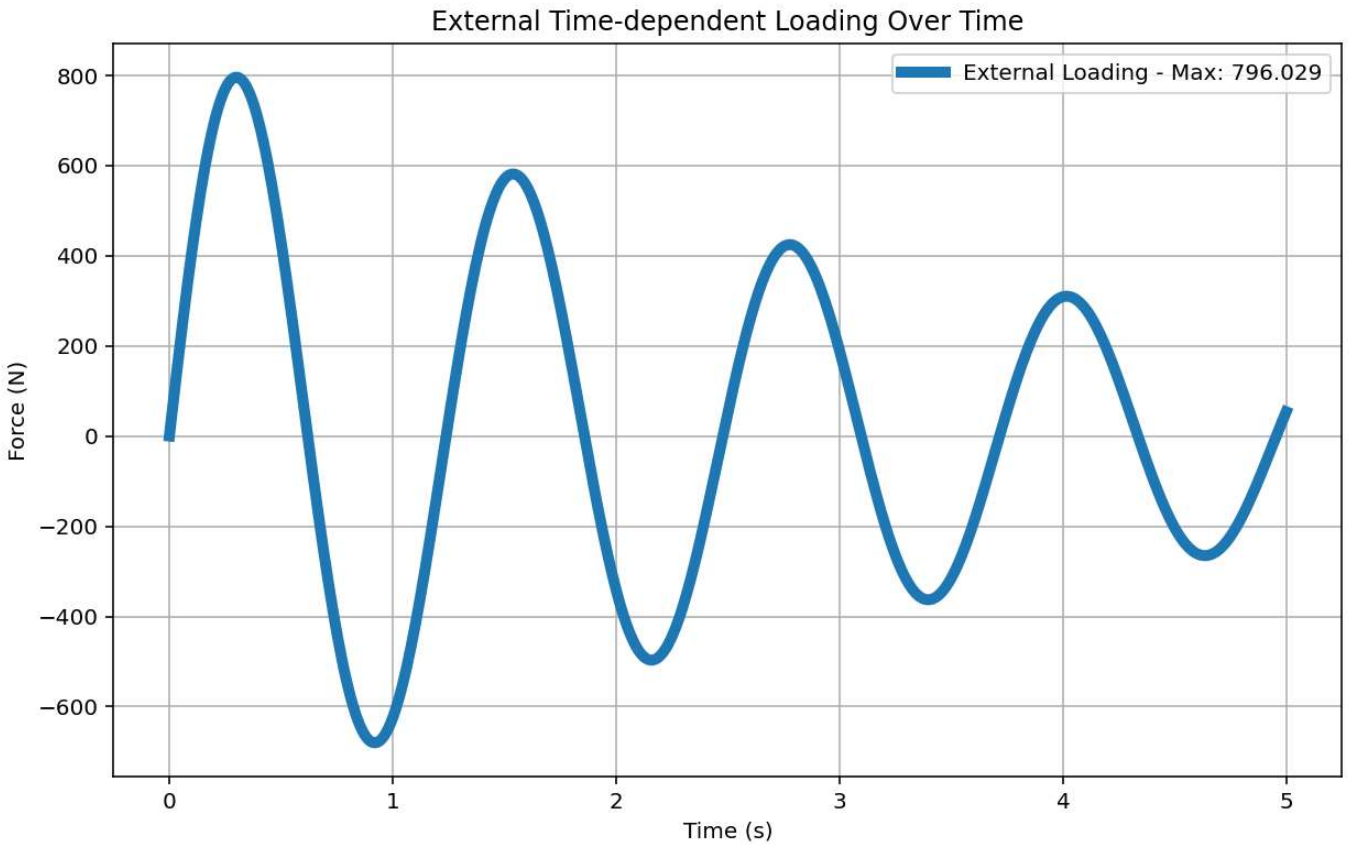


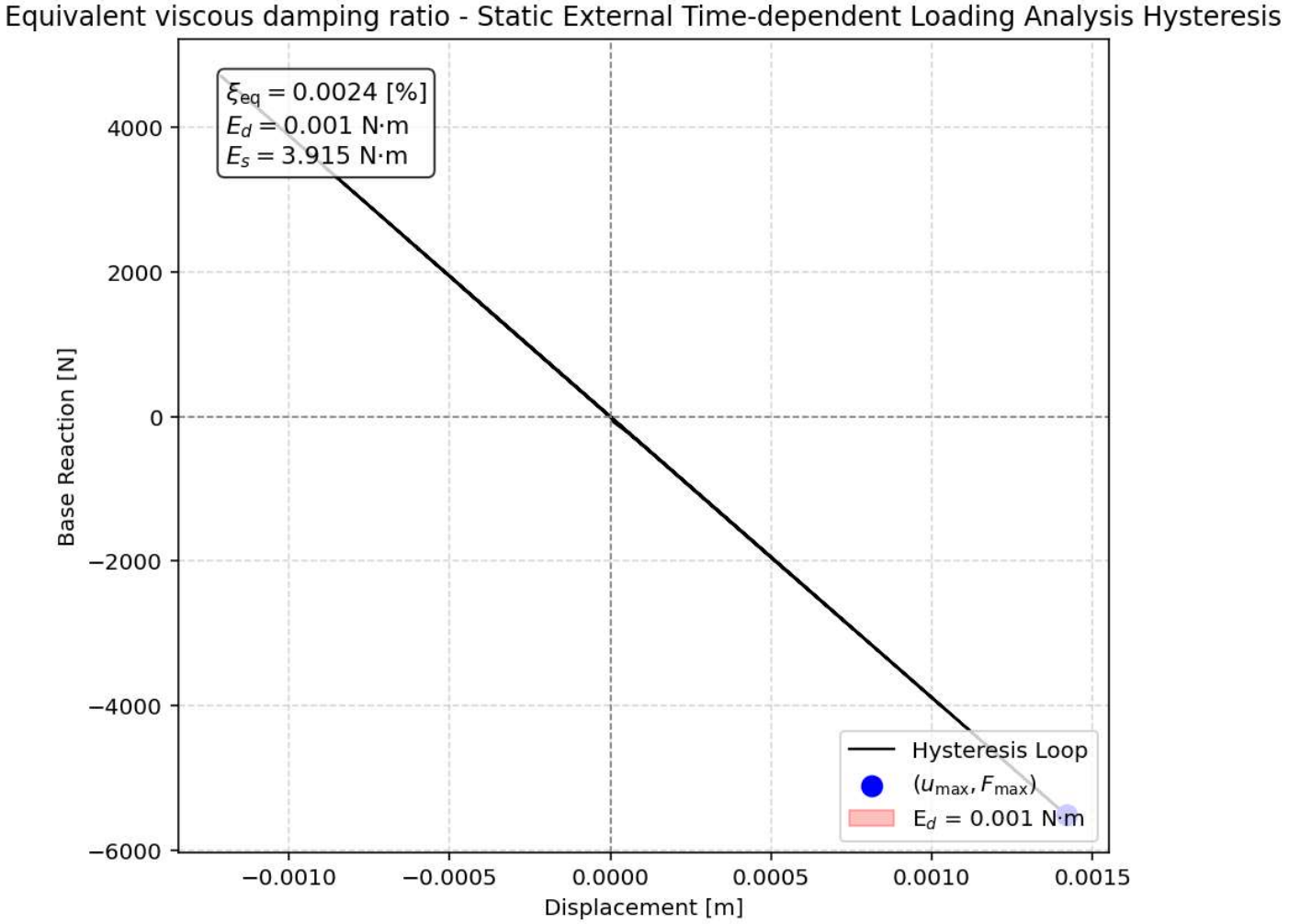
$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

CYCLIC DISPLACEMENT ANALYSIS

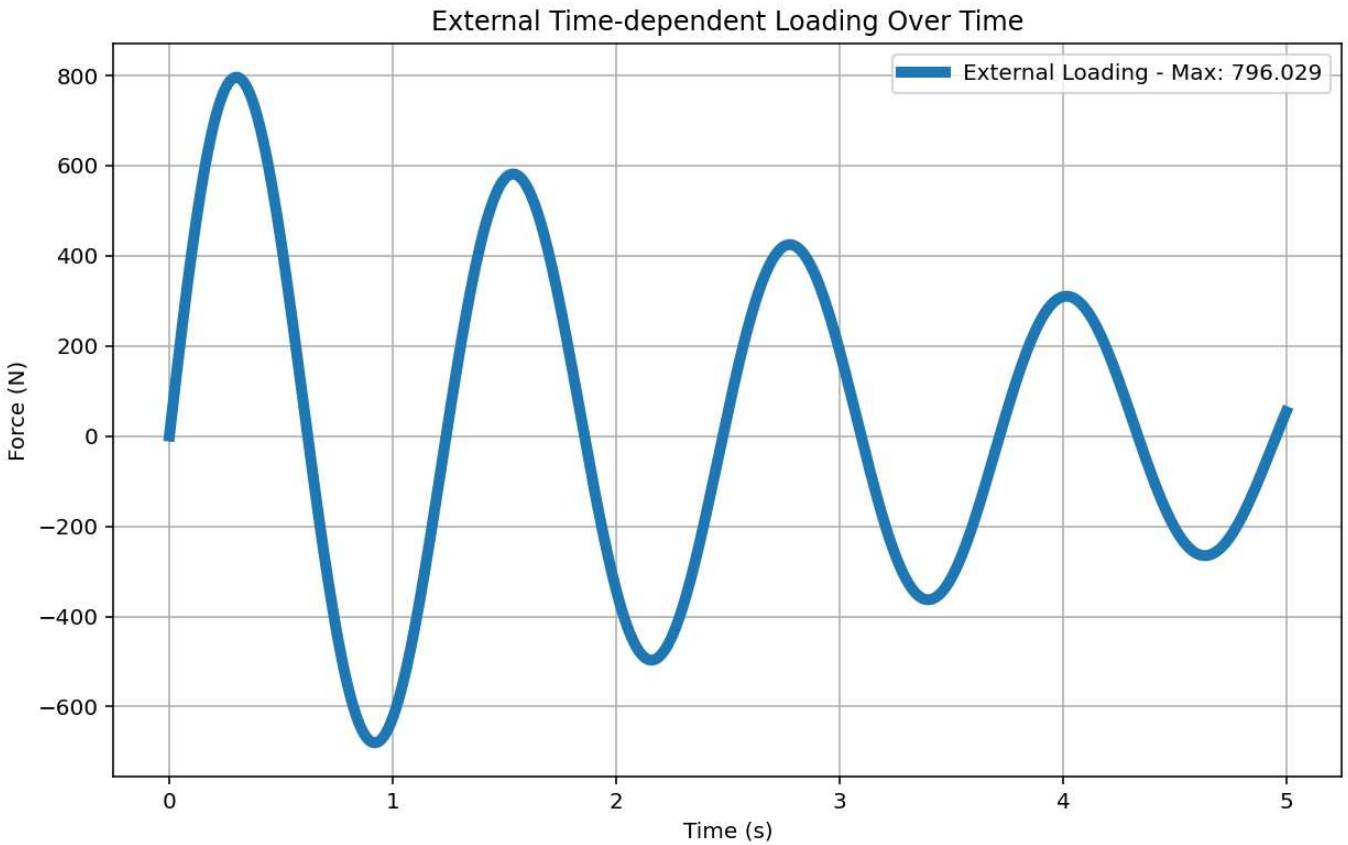

Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
$\xi_{eq} = 36.8611$ [%]
$E_d = 373026.809$ N·m
$E_s = 80530.695$ N·m
400000
200000
0
−200000
−400000
Base Reaction [N]
−0.3
−0.2
−0.1
0.0
0.1
0.2
0.3
Displacement [m]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 373026.809 N·m

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS

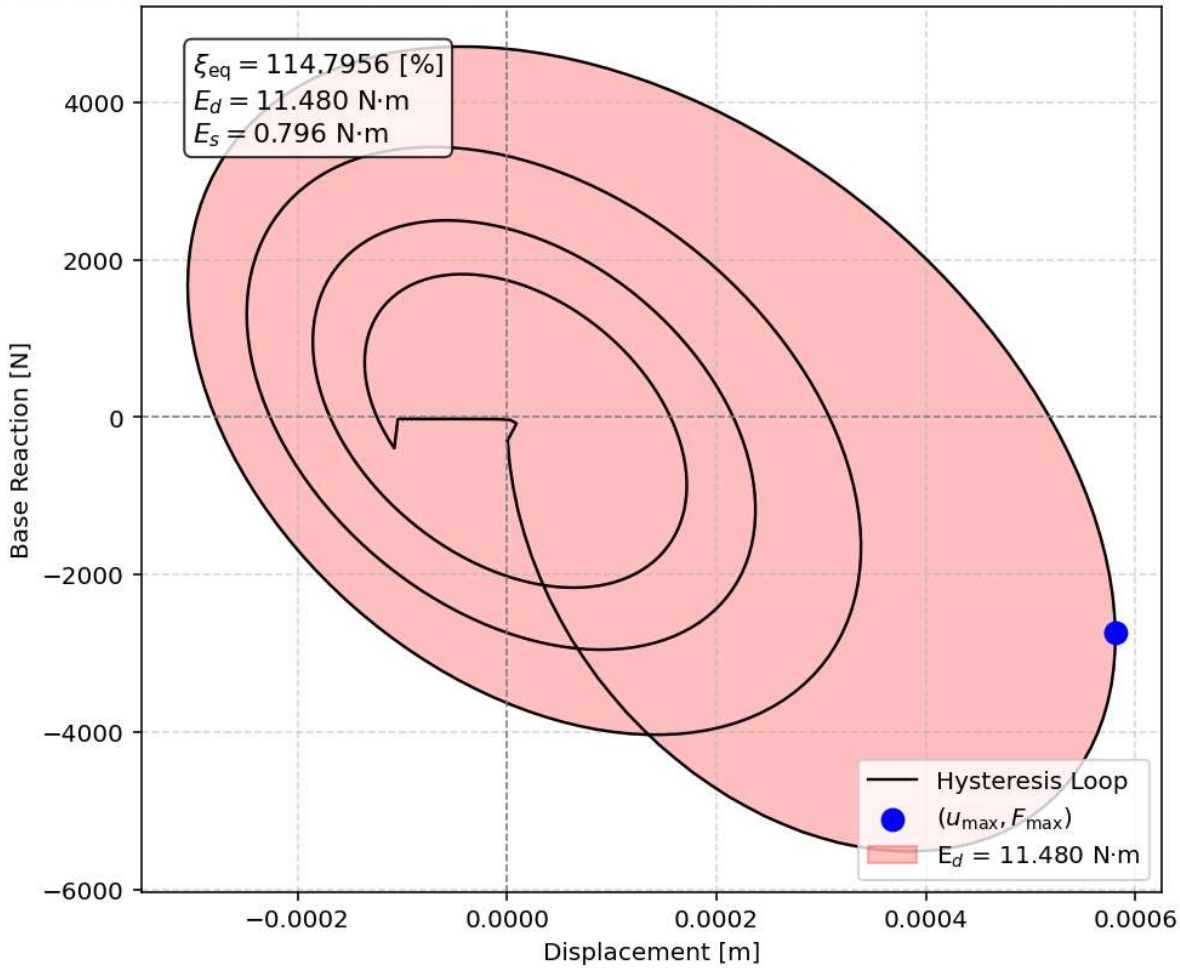

External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)


Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis
$\xi_{eq} = 0.0024$ [%]
$E_d = 0.001$ N·m
$E_s = 3.915$ N·m
Base Reaction [N]
Displacement [m]
4000
2000
0
−2000
−4000
−6000
−0.0010
−0.0005
0.0000
0.0005
0.0010
0.0015
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.001 N·m

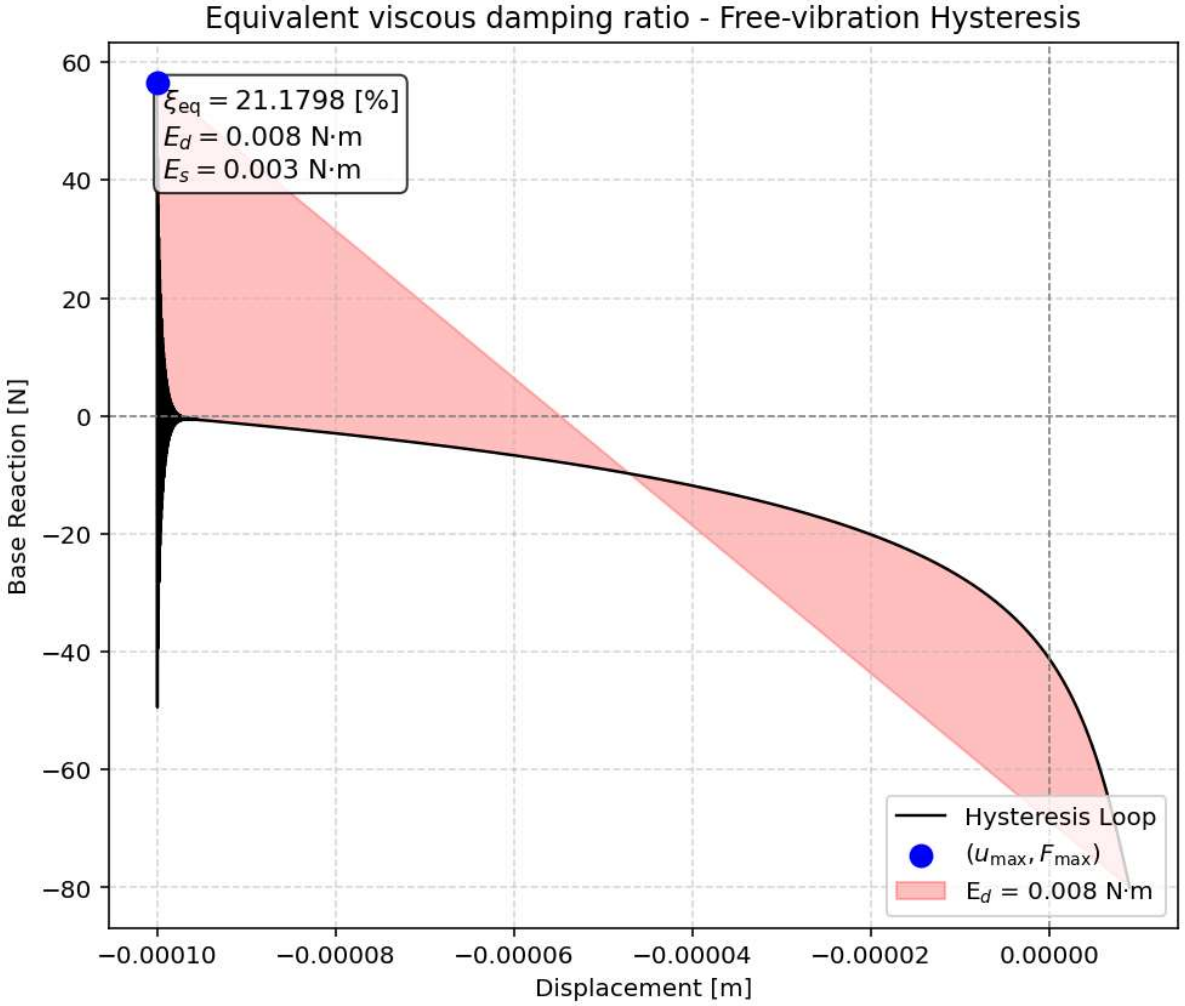
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS


External Time-dependent Loading Over Time
External Loading - Max: 796.029
Force (N)
800
600
400
200
0
−200
−400
−600
0
1
2
3
4
5
Time (s)

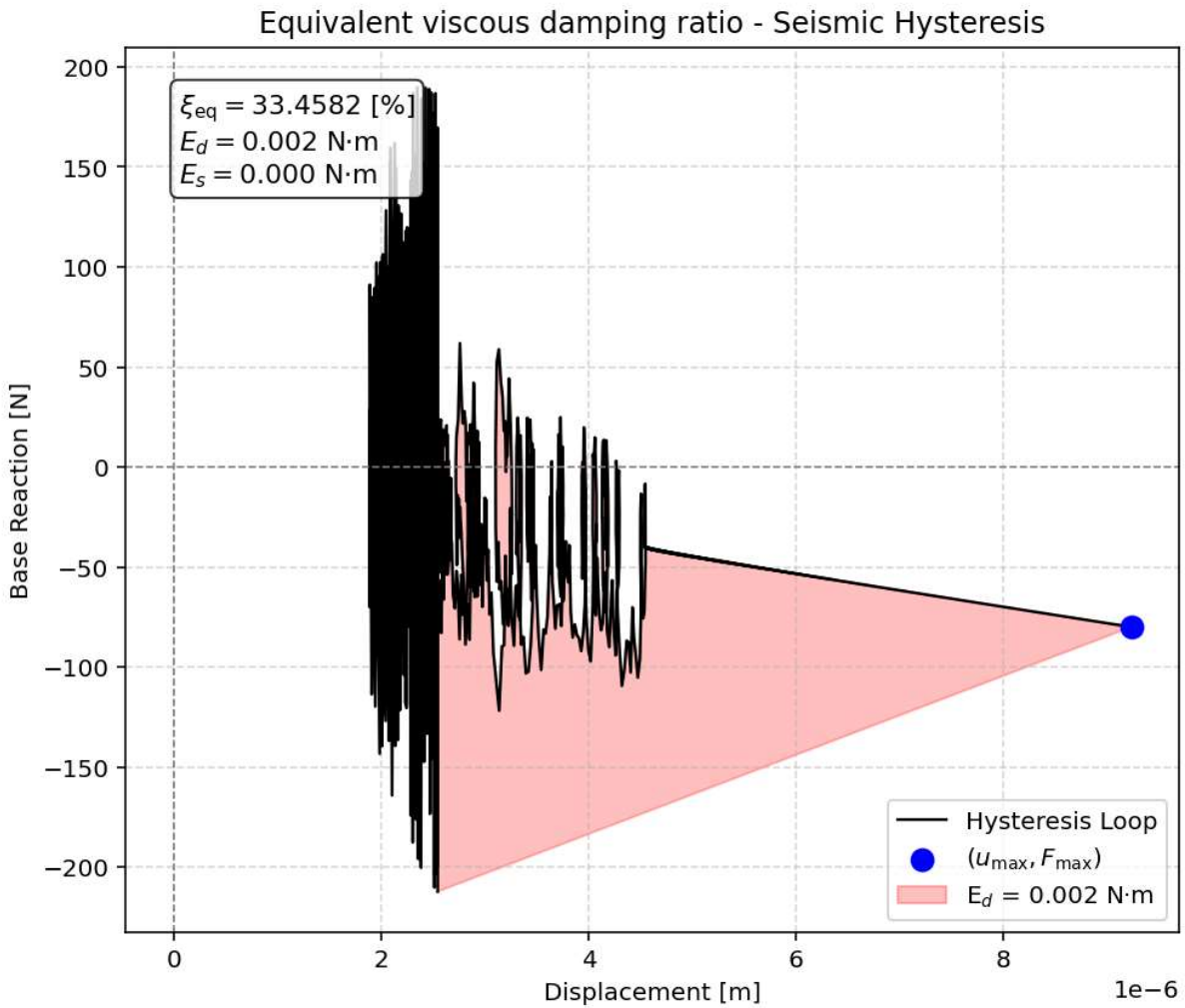
Equivalent viscous damping ratio - Static External Time-dependent Loading Analysis Hysteresis



FREE-VIBRATION ANALYSIS


Equivalent viscous damping ratio - Free-vibration Hysteresis
$\xi_{eq} = 21.1798$ [%]
$E_d = 0.008$ N·m
$E_s = 0.003$ N·m
Base Reaction [N]
Displacement [m]
60
40
20
0
−20
−40
−60
−80
−0.00010
−0.00008
−0.00006
−0.00004
−0.00002
0.00000
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.008 N·m

SEISMIC ANALYSIS


Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 33.4582$ [%]
$E_d = 0.002$ N·m
$E_s = 0.000$ N·m
Base Reaction [N]
Displacement [m]
1e−6
Hysteresis Loop
(u_{max}, F_{max})
E_d = 0.002 N·m